Beta Smoothing for Low-Label Predictive AI Evaluation

Lishuo Guo
Guizhou University of Finance and Economics
thebluegHostsss@mail.gufe.edu.cn

Abstract

Fixed Beta smoothing reduces small-budget probability error in a paired two-benchmark study for the Predictive AI Evaluation Competition. Candidate 989987 predicts $(k+1)/(n+2)$ from k successes in n revealed observations matching the target's visible subject and benchmark identity. It runs in the Python standard library with no trained parameters. Across 57 subject-benchmark pairs and 100 acquisition seeds, mean Brier area under the learning curve (ALC) is 0.219792, compared with 0.256956 for empirical frequency. We also examine a semantic-prior alternative and controlled transfer interventions. Intact cross-subject relations lose to matched shuffles, and explicit mean targets lose to matched controls across three training seeds. We describe the exact Beta artifact, its paired evaluation, and the training and inference procedures for the development alternatives.

1 Introduction

Predictive evaluation asks whether an AI system will succeed on an item before that system is executed. The competition instantiates this problem with visible subject and item attributes, held-out benchmark sources, and a small budget of revealed responses [1]. A predictor must be useful both at zero labels and after limited adaptation. The scoring rule rewards calibrated probabilities throughout this acquisition process, rather than classification decisions at a single threshold.

Two practical considerations motivate our approach. First, an empirical success frequency is unstable at the smallest budgets and can return extreme probabilities after a single observation. A fixed smoothing rule provides a transparent reference with almost no deployment burden. Second, text and visible subject attributes may provide a prior before outcomes from a new benchmark are available. Such a prior needs adaptation when its calibration shifts across sources.

Instance-level assessors and conditional benefits of learning across tasks and subjects are established topics [2]. Efficient aggregate benchmark evaluation [3], its extrapolation limits [4], and demand-profile representations [5] motivate our comparisons. We study how fixed smoothing, semantic priors, and sparse-label adaptation behave under benchmark-level cold start.

We provide a compact Beta predictor, the training and inference specification of a semantic alternative, and controlled comparisons of later transfer mechanisms. Each result is identified by its source panel and evaluation protocol. The candidate descriptions specify the exact packaged methods; the auxiliary studies examine subsequent research variants.

2 Task and evaluation metric

The input consists of a subject dictionary and an item dictionary. The subject describes a complete visible system configuration, including model name, provider, release and access dates, harness and version, reasoning effort, and additional settings. The item supplies its content, additional

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features, interactors, and an anonymous benchmark identifier. A target is one binary response, not a benchmark-wide aggregate.

For eligible subject-benchmark pairs, the published protocol partitions distinct items into acquisition and evaluation pools. Repeated responses of an item remain together. The budgets are $n \in \{0, 1, 3, 7, 15, 31\}$, and a single nested acquisition trajectory provides the revealed labels. Evaluation outcomes are not available to the predictor. Without an acquisition hook, the platform selects labels randomly [1].

Let B_n be the mean squared probability error at budget n. The normalized log-budget area is

$$\mathrm{ALC} = 0.1B_0 + 0.2B_1 + 0.2B_3 + 0.2B_7 + 0.2B_{15} + 0.1B_{31}. \tag{1}$$

These are trapezoidal weights on the equally spaced positions $\log_2(n+1) = 0, \ldots, 5$. The platform averages pair-level ALC equally. Lower is better; a constant probability of 0.5 gives Brier score 0.25. We state the aggregation rule for each offline study alongside its source panel.

Both implementations use revealed observations matching the complete visible subject dictionary and target benchmark identifier, retaining repeated observations. They predict from the supplied attributes and labels. The semantic runtime caches frozen feature computations and reconstructs calibration state for each call.

3 Candidate method and development comparison

3.1 Fixed Beta smoothing

For k successes among n matching revealed outcomes, the first candidate returns

$$\widehat{p}_{\mathrm{Beta}} = \frac{k+1}{n+2}. \tag{2}$$

This is the posterior predictive mean under a uniform Beta prior and a Bernoulli observation abstraction. It returns 0.5 with no labels and avoids probabilities of exactly zero or one after a finite number of observations. Its parameters are fixed rather than estimated from the reported diagnostic.

The rule pools matching observations into a group success rate and ignores differences in item difficulty. All zero-label targets receive 0.5. This makes it a simple reference for assessing the value of item features and information transferred from other sources.

The ZIP contains only `model.py` and `README.md`; it has no fitted state, network client, or `labeling.py`. The method uses the platform's default acquisition behavior. The baseline used in the early offline comparison is the official empirical frequency k/n, returning 0.5 when $n = 0$.

3.2 Frozen semantic representation and bilinear heads

The second candidate uses frozen `intfloat/multilingual-e5-small` (revision `614241f622f53c4eeff9890bdc4f31cfecc418b3`) and visible subject metadata. The local model manifest binds this identity to the packaged asset hashes. The exported runtime prepends `query:` to item content, tokenizes with a maximum length of 512, computes attention-mask mean pooling, and normalizes the 384-dimensional embedding. It uses tokenizer truncation. Explicit head-tail string splicing belongs to later research variants.

Each of the eight visible subject fields becomes a `field=value` token. A nonnegative 128-dimensional feature hash is normalized and concatenated with the text embedding. The concatenated vector is normalized again. For the resulting $x = [x_t; x_m] \in \mathbb{R}^{512}$, head h computes

$$z_h(x) = w_h^\top x + b_h + \frac{(A_h^\top x_t)^\top (B_h^\top x_m)}{\sqrt{8}}, \qquad p_0(x) = \frac{1}{5}\sum_{h=0}^{4} \sigma(z_h(x)). \tag{3}$$

The interaction rank is eight. Each head evaluates in float32, and its probabilities are averaged in float64. The runtime loads five frozen head states and a frozen encoder.

The full-development export records five head fits, seeds 0–4, on 76,455 records from 147 subject-benchmark pairs and ten sources. Each fit uses 256 updates, batch size 512, learning rate 0.01,

and weight decay 0.03. For a training record i in subject-source pair g, the target is its success fraction s_i/r_i. Sampling with replacement uses probability $r_i/(G\sum_{j\in g} r_j)$, where G is the number of training pairs. AdamW minimizes sampled BCE-with-logits, and gradients are clipped to norm 5. The sampling weights enter through record selection, with no second loss multiplier. Each head makes 131,072 draws. The feature builder uses the same query prefix and 512-token truncation recipe; the exporter checks its cache identity. Fit receipts retain the matrix and label hashes. Deployment weights use the full development corpus; offline assessment uses separate models fitted with each evaluation source excluded.

3.3 Call-local calibration and shrinkage

For each matching revealed observation j, let ℓ_j be the clipped logit of its frozen ensemble prior. A scalar calibration offset is obtained from

$$\widehat{\delta} = \arg\min_{\delta} \left\{ \sum_{j=1}^{n} \left[\log(1 + e^{\ell_j + \delta}) - y_j(\ell_j + \delta) \right] + 2\delta^2 \right\}. \tag{4}$$

The derivative is $\sum_j [\sigma(\ell_j + \delta) - y_j] + 4\delta$. Its derivative with respect to δ is at least four, giving a unique root. The implementation uses 160 bisection iterations over $[-n/4, n/4]$; with no observations, $\widehat{\delta} = 0$. Priors are clipped to $[10^{-6}, 1 - 10^{-6}]$ before taking logits.

For a target with prior $p_0(x)$, the final prediction is

$$\widehat{p}(x) = \tfrac{1}{2}\sigma(\mathrm{logit}(p_0(x)) + \widehat{\delta}) + \tfrac{1}{2}\frac{k+1}{n+2}. \tag{5}$$

At zero labels, the output is half the semantic prior plus 0.25. The offset, counts, and filtered observations are local to a prediction call. Evaluation adapts the scalar offset while keeping the trained heads fixed.

4 Data provenance and development design

Competition-specific data come from measurement-db revision `54074453da7b9911be0f48e4c560be547bc84d21`. The early smoothing diagnostic uses `real_webagents` and `researchcodebench`; the full semantic export uses the ten sources listed in Appendix A. The pretrained encoder is a separate model dependency. The accompanying bundle records data provenance, artifact identities, and upstream references.

The early diagnostic contains 57 eligible pairs across two binary-response benchmarks. Distinct items are split 50/50, repeated responses stay with their item, and one recorded response is acquired for each selected acquisition item. Methods share splits, acquired labels, and evaluation pools. The expanded diagnostic uses seeds 0–99, with the additional seeds declared before the extension. These seeds measure acquisition variability within the two benchmarks.

The semantic stability study uses ten development sources. It compares a five-training-seed ensemble with its recorded `map_reference` using a recipe selected during earlier development. The later studies use eight sources, 130 evaluation targets, and five evaluation splits under multiple evidence contexts. Within each study, the arms share the original acquisition trajectories.

Confirmation data have not been evaluated. The reserve inventory contains 20 source names whose eligibility, access, and independence require verification. MathConstraint remains subject to its original development prerequisite and family/condition grouping rules; the corresponding Berta expansion did not meet that prerequisite.

5 Results

5.1 Smoothing at small budgets

Table 1 reports the expanded two-benchmark diagnostic. Beta smoothing lowers ALC within each benchmark and by approximately 14.46% in the all-seed aggregate.

Table 1: Early offline diagnostic: mean Brier ALC, with equal pair weighting. The 100 seeds reuse the same source data. Smaller values are better.

Scope	Pairs	Constant	Empirical mean	Beta(1,1)
All 100 seeds	57	0.250000	0.256956	0.219792
Web-agent benchmark	26	0.250000	0.237090	0.205671
Research-code benchmark	31	0.250000	0.273617	0.231635
Additional seeds 20–99	57	0.250000	0.257300	0.219950

Table 2: Selected development contrasts on the reused eight-source panel. Each arm and control share their study's fixed protocol.

Within-study arm	Main ALC	Comparator ALC
Structural proxy vs. lexical proxy	0.214293649	0.214650344
Structural proxy vs. structural shuffle	0.214293649	0.214793944
Intact relation vs. pooled BCE	0.213814973	0.213896909
Intact relation vs. shuffled relation	0.213814973	0.213342562
Mean target, three heads, vs. pooled	0.214363446	0.214003448
Mean target, three heads, vs. shuffled	0.214363446	0.213832837

5.2 Semantic-prior development evidence

In the ten-source stability study, the five-head ensemble obtains ALC 0.217693569, compared with 0.220140833 for its fixed reference. It improves on seven sources and regresses on three. The result measures the selected development recipe on this ten-source panel.

The export and text-only compatibility repair passed their recorded artifact checks and independent engineering review. Their exact implementation is described in the reproducibility section.

5.3 Controlled transfer diagnostics

Table 2 reports paired contrasts within the later studies. Prior construction and head count vary between studies, so comparisons use the matched arm and control in each row.

The fixed structural proxy passes its structural diagnostic and improves over the specified head-tail comparator by 0.000741693, below the predefined material threshold of 0.002. A secondary linear structural construction fails the diagnostic. Block removal and renormalization jointly change information and kernel geometry. The comparison therefore evaluates this feature construction. Its inexpensive proxies are simpler than the rubric-based representation in [5].

The intact same-content cross-subject auxiliary improves slightly over pooled BCE but loses to its matched shuffled control. In this single-training-seed comparison, intact pairing offers no advantage over the shuffle. The shuffle preserves source, visible-subject, and trial-count strata, providing a control for information retained within those groups.

The explicit leave-self-out stratum-mean auxiliary was tested with three training seeds. Its ensemble loses to both matched three-head controls, and each individual mean model loses to the corresponding single-seed controls. The experiment comprises 72 normal fits, two excluded-source poison refits, and 105,300 saved score rows. The squared-loss expectation identity holds for fixed parameters; the experiment tests what happens after optimization. The mean-target replacement fails its original gate, and the shuffled arm remains a control.

5.4 Historical platform observations

The 13 October records show Beta submission `982149` as `Finished` with displayed Brier ALC 0.206299, and semantic submission `982956` as `Finished` with Brier ALC 0.2112249269. The later Beta run `989987` is also `Finished`, with displayed Brier ALC 0.181180. Its downloaded submission ZIP matches the Beta artifact hash in Appendix A. The evaluations used different formative samples, so these scores describe their individual runs.

The semantic receipt also contains a missing `PreTrainedModel` import notice and an unspecified runtime error. Their causes remain unresolved. Final summative evaluation on a common hidden subset is the relevant comparison between entries.

6 Implementation and reproducibility

The semantic ZIP bundles the encoder, tokenizer, five head states, manifests, inference code, dependency declarations, and concurrency configuration. It has 16 members and size 283,976,581 bytes; the Beta ZIP has two members and size 1,267 bytes. Read-only audits verify archive hashes, member CRCs, and equality between packaged `model.py` files and their reviewed staged sources. The reports describe these fixed artifacts.

The semantic runtime verifies asset hashes before loading. It uses local CPU float32 inference with two Torch threads, prediction concurrency one, and remote code disabled. The compatibility variant disables the optional `torchvision` availability flag before lazy text-model imports. It uses a private interface pinned to Transformers 4.57.6 in a dedicated text-only process.

Differential tests reproduced the optional-backend failure on the E5 path and found zero feature and prior discrepancy between repaired and clean execution on six records. A separate matrix-level replay checks the original export. The official ZIP checker and tiny smoke test exercise the recorded interface cases. Broader long-input and dependency-environment coverage remains untested.

The first 20-seed smoothing computation was independently reconstructed without importing the candidate or evaluator modules. All 20,520 metric cells and 1,140 split manifests matched within 1.11×10^{-16}. The 100-seed extension contains 102,600 metric cells and 5,700 split records. Detailed receipts accompany the reproduction guide.

The public repository provides the exact Beta predictor and standalone Beta replay. A separate five-scope bundle recomputes development predictions, observations, and gates (Appendix B). The 155 fixed fits reproduce the original parameters using frozen embeddings; fresh encoding remains unexecuted.

7 Limitations and next steps

Beta smoothing ignores item difficulty. Semantic priors provide item-specific predictions but can be miscalibrated on new sources; scalar adaptation adjusts a group offset while leaving item-dependent errors. Both packaged candidates use random acquisition.

The offline results describe finite development panels with different sources, selection histories, and aggregation rules. Related subjects, repeated responses, and shared source data induce dependence. Acquisition seeds measure variability within these panels. Estimating performance across new sources requires a separate frozen panel and a suitable uncertainty analysis.

The next comparison should match pooled, intact, shuffled, and explicit-mean objectives on representation, architecture, head count, training schedule, and inference. It should fix its primary contrast and source-level uncertainty analysis before evaluating eligible untouched sources. This would test whether the present mechanism comparisons persist beyond development.

Disclosure. AI coding and research assistants supported implementation, verification, and manuscript preparation. Experimental labels came from the recorded measurement database; no paid prediction API generated them. Development and reviewer-service costs have not been quantified.

8 Conclusion

Fixed Beta smoothing improves on empirical frequency in both benchmarks of the paired low-label diagnostic. Candidate 989987 implements this rule in a compact, call-local predictor. The semantic alternative adds item-specific prior information, while the later controls reveal where proposed transfer mechanisms fail to outperform matched alternatives. These results establish a simple reference and concrete comparisons for subsequent evaluation on new sources.

References

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A Data and artifact identities

The full semantic export uses `algotune`, `appworld`, `benger`, `ceval`, `critic_discernment_game`, `devbench`, `financebench`, `helm_thaiexam`, `hle`, and `igakuqa`. The Beta diagnostic uses `real_webagents` and `researchcodebench`; the Beta rule remains fixed throughout. The evidence map records source versions, encoder identity, and licensing metadata.

Table 3: Existing candidate identity. Hashes refer to exact ZIP bytes.

Candidate	SHA-256
Beta(1,1), ID 989987	b5fc15fee6c5af12b2cbaa79ac4828ac72a8765a2d7ee18891ac674994724e03
V6 compatibility, ID 982956	62c819343ab38d77c77da80e9197eb998c8b33b04f8239239652058a7c24ecd2

B Public reproduction

The exact Beta predictor and standalone replay are publicly available under the MIT license at https://github.com/THEBLUEGHOSTSSSS/paiec-beta-reproducibility. A clean-directory replay executes the submitted predictor across all 100 seeds; all 102,600 metric rows agree with the original audit within 10^{-12} and all 5,700 split records match exactly. A separate five-scope bundle contains the code and a hash-bound derived reference archive; it is publicly available at https://github.com/THEBLUEGHOSTSSSS/paiec-beta-reproducibility/releases/tag/reproduction-20261031. Local validation passed; independent model review remains incomplete because of transport errors. A separate clean-directory execution reproduced 39,690 semantic scoring cells and 315,900 transfer score rows; maximum score discrepancy was 1.67×10^{-16}, with unchanged decision gates. All 155 fixed head fits reproduced the original parameters. Raw tables and pinned encoder assets are obtained directly from their upstream sources. Fresh encoding and hosted deployment were not rerun.

C Evidence map

Report-to-artifact mappings and hashes are retained in `paiec/paper_20261029/evidence/evidence.json`. The reproduction bundle includes source and input manifests, executed replay and training receipts, and original decision-gate comparisons. Historical platform observations remain separate from these local checks.